

SentinelX

6. Data Sources

SentinelX uses two independent AI components, each configured on different data and deployed in different places.

Bi-LSTM — On-Device Threat / Non-Threat Language Classification

The Bi-LSTM model was trained on a labeled corpus of threat and non-threat statements sourced from Kaggle: “Threat and Non-Threat” dataset — <https://www.kaggle.com/datasets/iorhentomna/threat-and-non-threat> — totaling over 100,000 labeled examples. This dataset was used specifically to teach the model to distinguish threatening language patterns from non-threatening ones at the statement level. After training, the model was converted into a lightweight on-device format and bundled into the mobile app, so it never requires a server or network connection to classify transcribed speech from the Detect Threat feature.

Dataset split:

- 70% training
- 20% validation
- 10% testing

This produced 96% test accuracy on threat vs. non-threat classification.

Gemini API — Uploaded Evidence Analysis

Gemini is used at inference time only, to analyze evidence — video, audio, or image files — uploaded by the user through the Safe Chat “Upload Evidence” step. It is not trained or fine-tuned on any of our data; it receives the uploaded evidence file for a single report and returns a contextual analysis of that evidence for the analyst to review, and only after an analyst clicks [Analyze Report]. No training data is involved on our side for this component.

Why no training corpus applies to Gemini: because Gemini analyzes real but user-submitted evidence at the point of analyst review rather than being trained by us, there is no training dataset to disclose for it — only the data flow itself, which is the user’s own uploaded evidence, sent only after explicit analyst authorization. Gemini’s own underlying training data is Google’s and outside

the scope of what SentinelX is responsible for disclosing, since no SentinelX user data is used to train Gemini itself.

On Real Case Data and Raw Audio

In line with the challenge brief's requirement, SentinelX never used real survivor, victim, or confidential case data to train any model — the Bi-LSTM's training data is the named Kaggle source above. The only “real” data in the system is a user's own report, evidence upload, or Detect Threat transcript snippet about their own situation — analyzed live or reviewed by an analyst, never used for training. Raw audio captured by Detect Threat is never retained at all, by design.